

Assignment-1

1. What is Full Stack Development?

Full Stack Development is the development of both the frontend and backend of a web application. A Full Stack Developer designs the user interface, develops server-side logic, manages databases, and integrates all components of the application.

2. What is Frontend?

Frontend is the part of a website that users can see and interact with. Common frontend technologies include HTML, CSS, JavaScript, React, Angular, and Vue.js.

3. What is Backend?

Backend is the server-side part of a web application that handles data processing and business logic. Its main responsibilities include managing databases, processing requests, handling authentication, and creating APIs.

4. What is database?

A database is an organized collection of data used to store and manage information. Websites need databases to save and retrieve data efficiently. Examples include user details, login credentials, product information, and order records.

5. What is DNS?

DNS (Domain Name System) translates domain names into IP addresses, helping users access websites using easy-to-remember names instead of numerical addresses.

6. What is an IP Address?

An IP (Internet Protocol) address is a unique numerical identifier assigned to a device on a network. IP addresses are important because they enable devices to communicate and exchange data over the internet.

7. What is Client-Server Architecture?

Client-Server Architecture is a model in which the client (such as a web browser) sends requests for services or data, and the server processes those requests and returns the required response. This communication allows users to access websites and web applications.

8. What is an API?

An API (Application Programming Interface) is a set of rules that allows different software applications to communicate with each other. In web applications, APIs help the frontend exchange data with the backend, enabling functions such as user login, fetching data, and processing requests.

9. What are HTTP Methods?

HTTP methods are used to define the type of action performed on a web server.

GET: Retrieves data from the server.

POST: Sends new data to the server.

PUT: Updates existing data on the server.

DELETE: Removes data from the server.

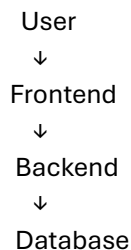
10. What are Developer Tools?

Developer Tools are built-in browser tools used to inspect, debug, and test web pages.

Three tabs in Developer Tools and their purposes:

- Elements: Used to inspect and edit HTML and CSS.
- Console: Displays errors and allows JavaScript execution.
- Network: Monitors requests and responses between the browser and server.

11. Full Stack Architecture Diagram



Short Description of Each Layer:

- User: The person who interacts with the website or application.
- Frontend: The visual part of the application that users see and use (built using HTML, CSS, JavaScript, etc.).
- Backend: The server-side component that processes requests, applies business logic, and communicates with the database.
- Database: Stores and manages application data such as user information, products, and transactions.

12. Complete Website Working Flow

Explanation of Each Step:

- User: The user enters a website address or clicks a link.
- Browser: The browser sends a request to access the website.
- DNS: DNS converts the domain name into the server's IP address.
- Server: The server receives and processes the request.
- Database: The database provides the required data to the server.
- Server: The server prepares and sends the response back.

- Browser: The browser receives the response and renders the webpage.
- Website Displayed: The requested website is shown to the user.

13. Student Registration System

Frontend: The student enters details in the registration form and submits it.

Backend: The server validates the information and processes the registration request.

Database: The student's details are stored and managed in the database.

14. Amazon Search Scenario

When a user searches for "Laptop" on Amazon, the Browser sends the search request to the Server. The Server checks the Database for laptops matching the search term.

The Database returns the relevant product details to the Server. The Server then sends a Response containing the search results back to the Browser, which displays them to the user.

15. Website Observation Activity

Website Name: Google (www.google.com)

Observation from the Elements Tab: The webpage structure is created using HTML elements such as <div>, <form>, and <input> tags.

Observation from the Network Tab: Multiple requests are sent to load resources like images, CSS files, and JavaScript files.

Observation from the Console Tab: The Console displays JavaScript messages, errors, and warnings related to the webpage.

16. How does a website work?

When a user enters a URL, the browser sends a request to the server through DNS. The server processes the request, retrieves the required data from the database, and sends a response back to the browser, which displays the webpage on the screen.

17. Difference between Frontend, Backend, and Database (Example: Instagram)

Frontend: The Instagram app interface where users view posts, stories, and profiles.

Backend: Processes actions such as uploading photos, liking posts, and managing user requests.

Database: Stores user accounts, posts, comments, likes, and messages.